The Direct and Indirect Effects of Genetics and Education

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The Direct and Indirect Effects of Genetics and Education

Senan Hogan-Hennessy, 14 November 2024.

Notes for LWIPS:

This project integrates data, methods from other fields to labour econ topics.

1. New genetic measures (from genetics)
2. An explicit mediation analysis (methods from statistics/epidemiology)
3. Aims to use these in way that is credible in labour econ, natural experiment setting.

Appreciate comments on what is confusing + unclear from an econ audience, given these unconventional approaches.

Note: Explains an overlapping research design approach for causal effects using UK Biobank data; causal results with these data are not ready here.

Introduction

Nature vs Nurture

1. Genetics measure nature (initial endowment)
2. Genes for social outcomes
3. Deeper understanding of social outcomes, from genetics.

Introduction

"**The difference of natural talents in different men**, is, in reality, much less than we are aware of [. . .]. The difference between the most dissimilar characters, between a philosopher and a common street porter, seems to arise **not so much from nature, as from habit, custom, and education**." *Adam Smith (1776), The Wealth of Nations, Volume 1 p. 16.*

Nature vs Nurture

1. Genetics measure nature (initial endowment)
2. Find genes associated with outcomes
3. Deeper understanding of social outcomes, from genetics.

Introduction

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Nature(?) vs Nurture

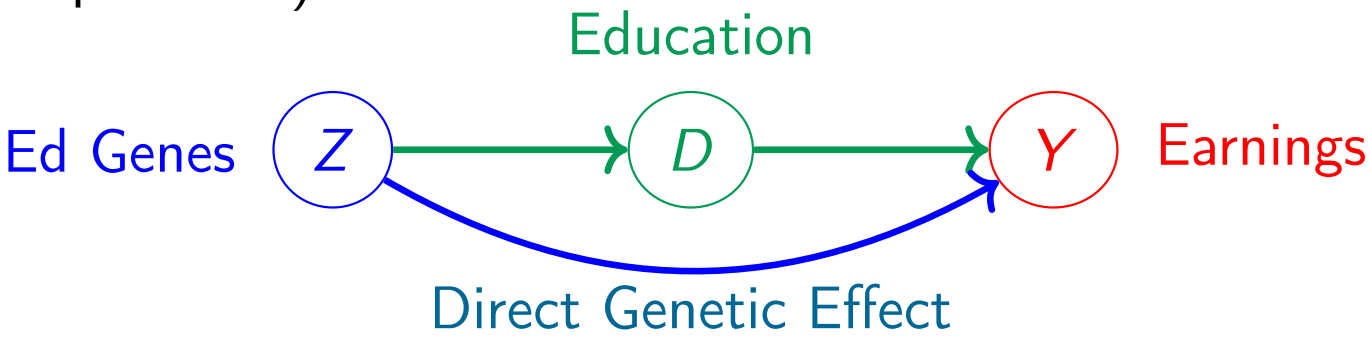
1. Genetics measure one component of nature (at most...)
2. Genes associated with outcomes are not the entire story
3. Genetics gives new findings, **with caveats.**

Introduction

How do genes associated with education affect labour outcomes?

1. Causal design for random variation in a genetic measure for education.

2. Decompose mechanisms by a mediation analysis (with an overlapping natural experiment).

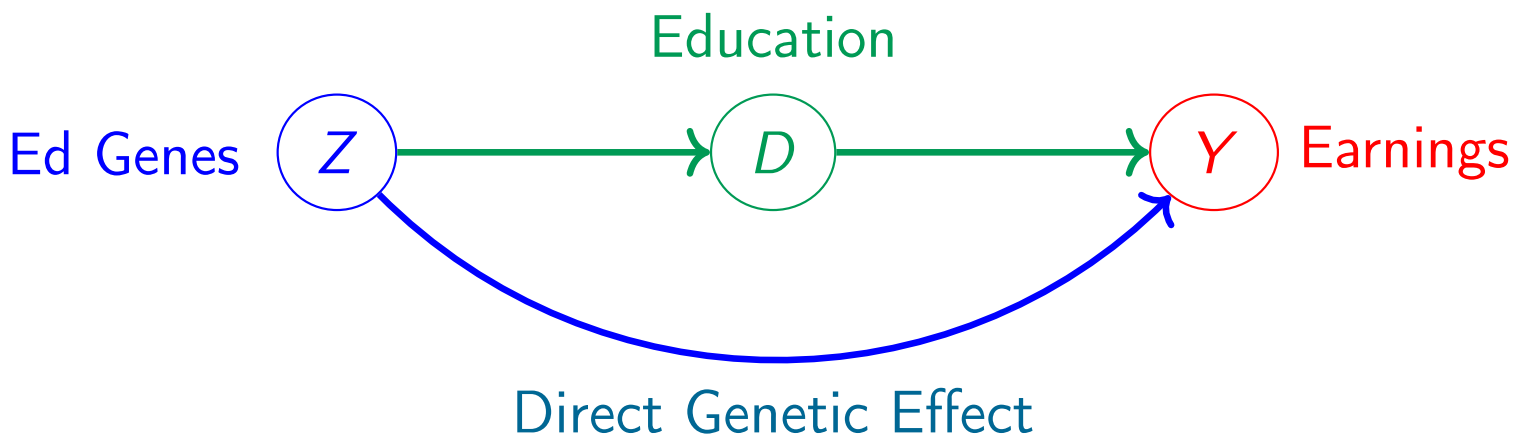


3. Finding that genes associated with education do/not affect outcomes **directly or indirectly,** inferring they do/not measure intelligence, aptitude, or other factors.

Introduction

How do genes associated with education affect labour outcomes?

"The evidence suggests fluid intelligence and self-control partly mediate the relationship between the PGI [genetic measure] and education. Genetic effects are economically meaningful, comparable in size to effects of family environment." *Abstract, Carvalho (2024, JPE Micro).*



- Genetic effects could be entirely returns to education
- Disputing claims that genetic measures pick up "fluid intelligence."
- Ratio, direct / indirect, is unclear

Introduction

How do genes associated with education affect labour outcomes?

Introduction

1. **Genetics & Education**
   Data: ~~UK Biobank~~ Health & Retirement Study (HRS)

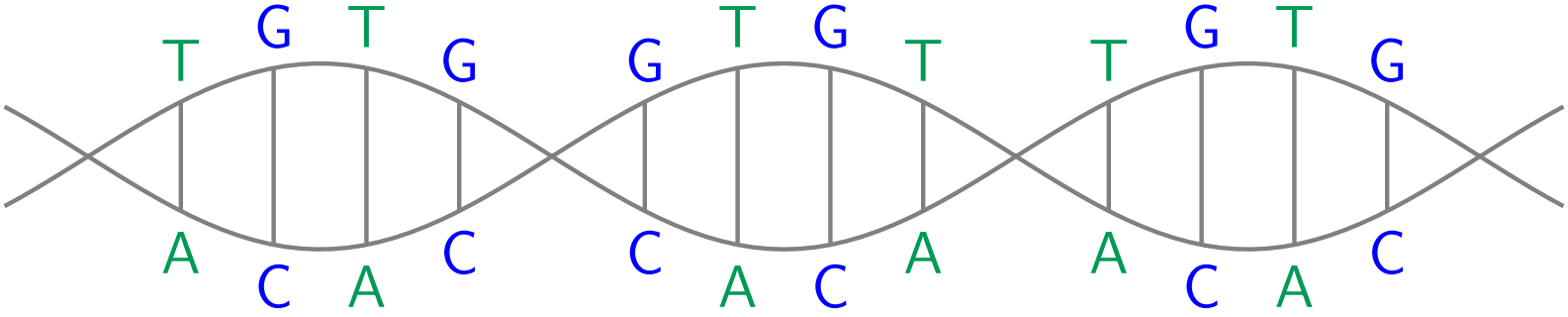
2. **Genetic Effects**
   Mendelian Independent Assortment

3. **Direct and Indirect Effects**
   Ed PGI's Effects Through Education

Genetics: a Primer

- The human genome is a 3 billion-long sequence of base pairs
- $\approx 0.1\%$ these pairs differ between two average people

Person 1's DNA:



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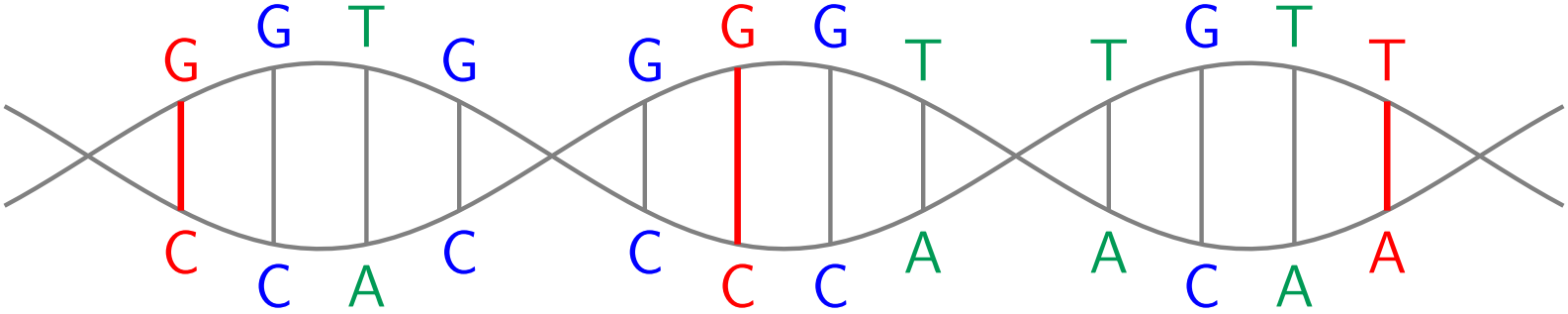

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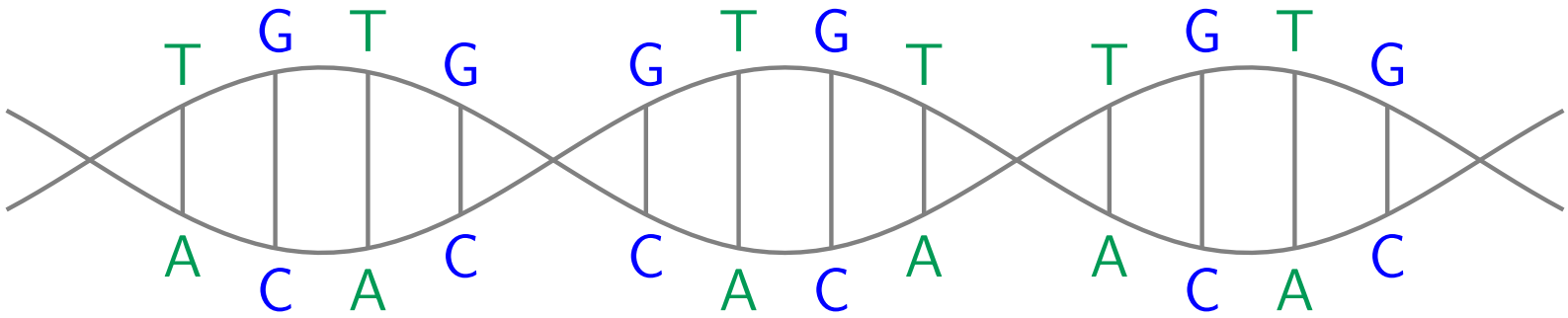

Gene:

Protein: X

Outcome

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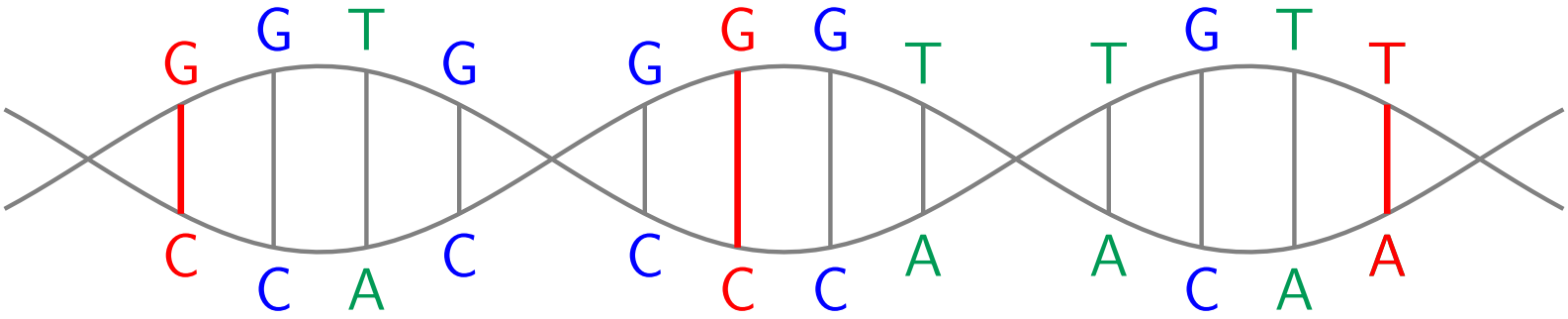

Gene:

Protein: X

Different outcome.

Moderated by external factors

"gene–environment interactions."

Polygenic Index & Education (Ed PGI)

"A large-scale genetic association analysis of educational attainment in a sample of approximately 1.1 million individuals and identify **1,271 independent genome-wide-significant SNPs... generating a Polygenic Index** that explains 11–13% of the variance in educational attainment and 7–10% of the variance in cognitive performance." *Abstract — Lee et al. (2018, Nature Genetics)*

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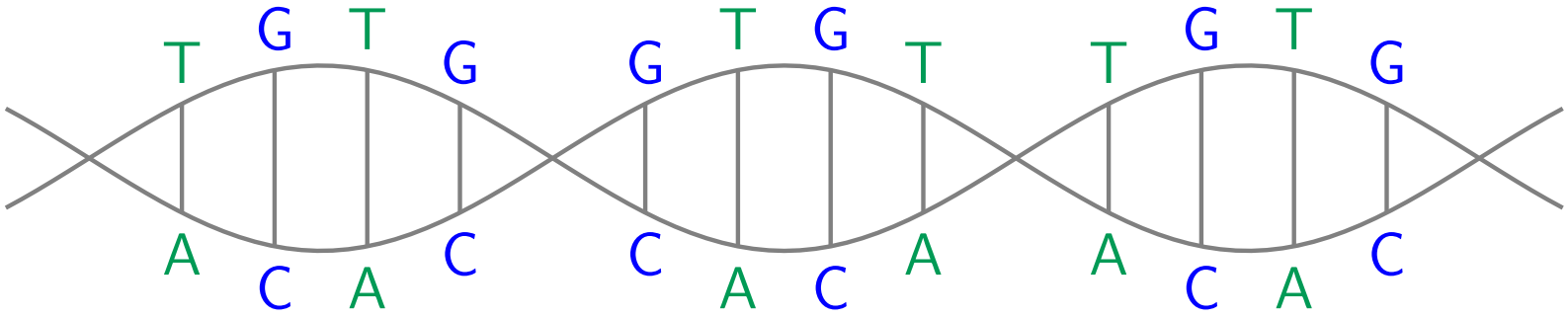


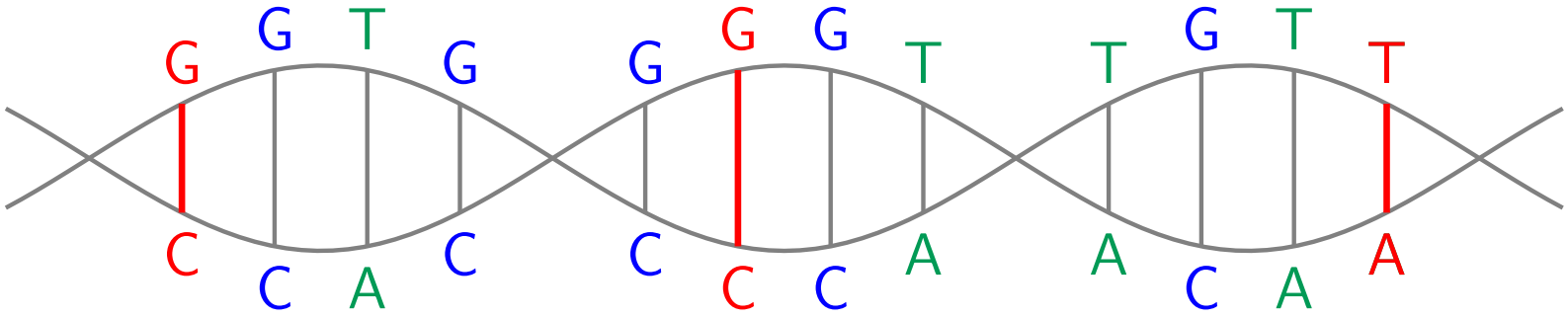
$\geq$1,271 Different Genes:

$\geq$1,271 Different Proteins: X

Education Attainment

Polygenic Index & Education (Ed PGI)

Person 2's DNA:



$\geq$1,271 Different Genes:

$\downarrow$

$\geq$1,271 Different Proteins: $\boxed{X}$

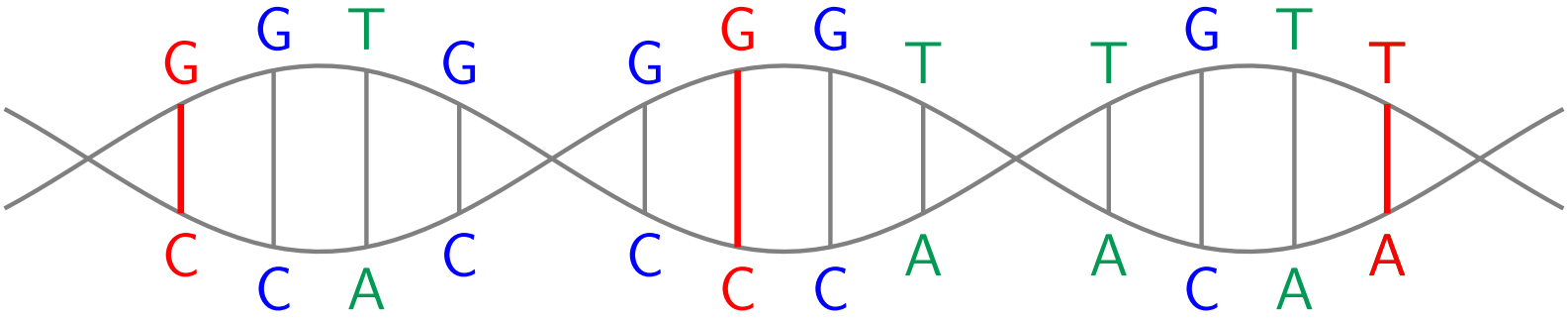
$\downarrow$

Education Attainment

- In practice, we do not know exact location of relevant gene regions
- Nor the associated protein + what they do

Polygenic Index & Education (Ed PGI)

Person 2's DNA:



Calculate Polygenic Index (PGI):

1. Identify mutation regions (1,271 for Ed PGI)
2. Calculate weights for each
3. Assign a 1-dim genetic measure, summarising + dimension-reducing this topic

Education Attainment

- In practice, we do not know exact location of relevant gene regions
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Polygenic Index & Education (Ed PGI)

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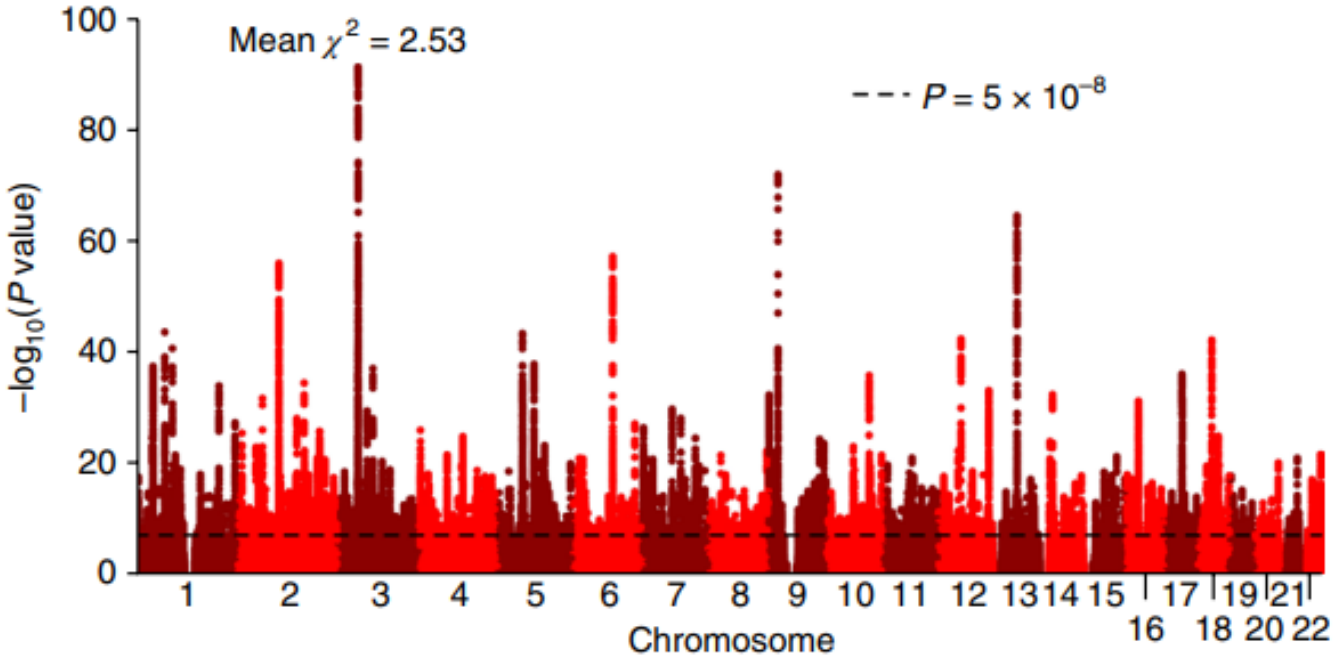
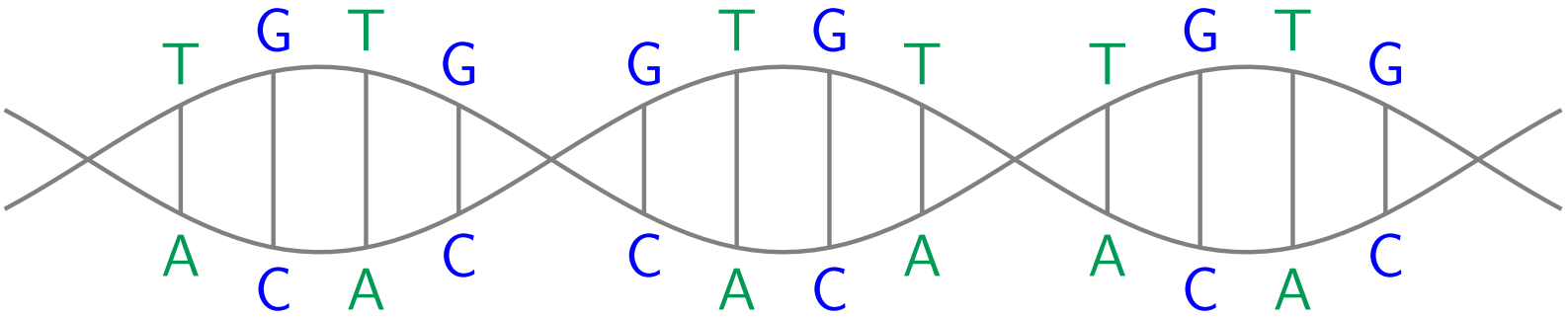


Fig. 1 | Manhattan Plot for GWAS of EduYears. The *P* value and mean

Polygenic Index & Education (Ed PGI)

Education Polygenic Index (Ed PGI):
A weighted sum of single letter gene changes predicting years of education.

Person 1's DNA:



Weights: w_1 0 0 0 0 w_2 0 0 0 0 0 w_3

Person 1's Ed PGI:

$$Z = \sum_{j \in J} w_j \mathbb{1}\{\text{one-letter change at } j\} = 0$$

Polygenic Index & Education (Ed PGI)

Education Polygenic Index (Ed PGI):
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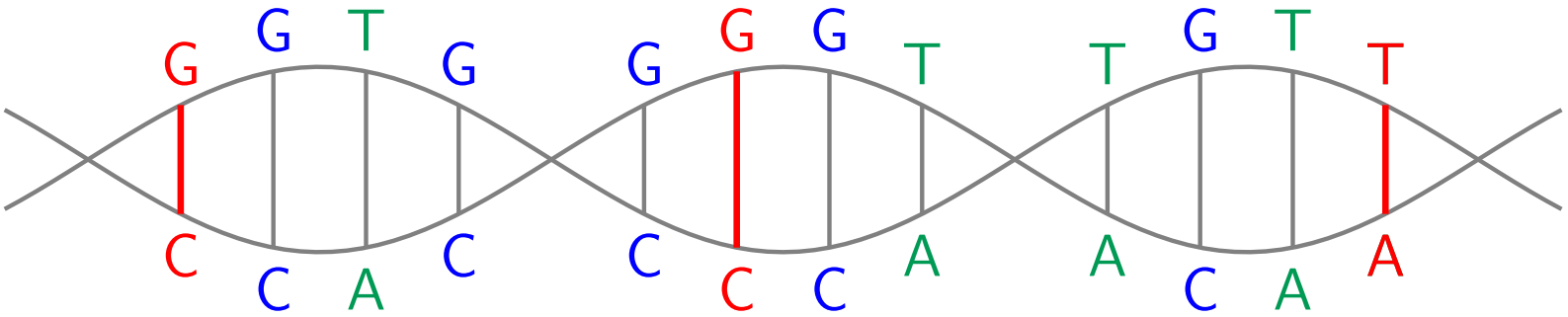
Person 2's DNA:



Weights: w_1 0 0 0 0 w_2 0 0 0 0 0 w_3

Person 2's Ed PGI:

$$Z = \sum_{j \in J} w_j \mathbb{1}\{\text{one-letter change at } j\} = w_1 + w_2 + w_3$$

Note: Humans have two copies of each chromosome, so indicator is 0, 1, 2 corresponding to how many copies one has.

Education Polygenic Index (Ed PGI)

Previous economics + science research shows Ed PGI is strongly associated with multiple labour market outcomes:

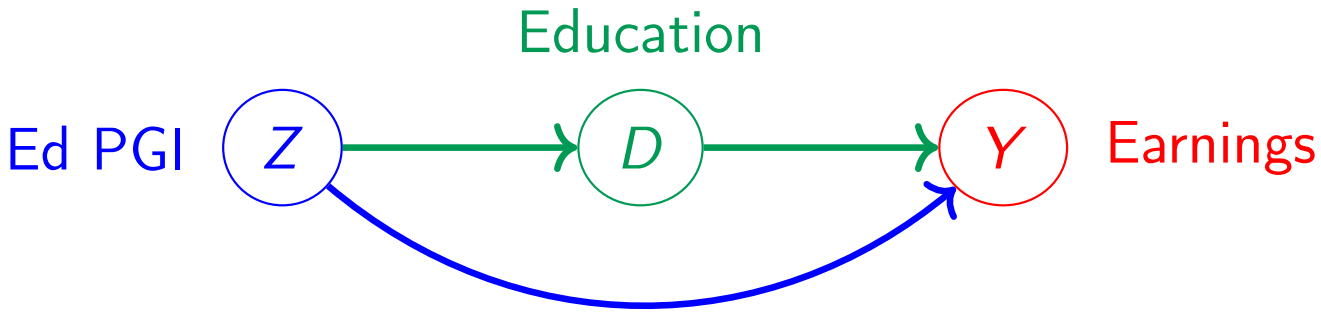
1.	Retirement wealth	(Barth et al., 2020)
2.	Life-cycle income	(Barth et al., 2022)
3.	Longevity	(Marioni et al., 2016)
4.	Primary school performance	(Aagaard-Houmark et al., 2022)
5.	Parental–child care (nurture)	(Aagaard-Houmark et al., 2024)
6.	Socioeconomic Status	(Carvalho, 2024)
7.	...	
8.	Many in-progress	(Forthcoming)

Open question: What are the mechanisms behind Ed PGI's effects?

- Is education the **only way** it is related to outcomes?
- Is there a **direct genetic relationship**, absent returns to education?

Education Polygenic Index (Ed PGI)

I look at how Ed PGI **causally affects** labour market earnings, and **model how much this relationship is because of the education channel.**



Total Effect	=	Indirect Effect	+	Direct Effect
(Ed PGI → Earnings)		(Education)		(Genetic)

Contributions:

- Connect new genetic measures to classic labour econ topics
- Mediation methods in a natural experiment setting
- **Test:** *"Genetic measures correlated with education affect earnings* ***beyond just education****" — Barth Papageorge Thom (2020).*

Data: UK Biobank

This is **the** dataset that genetics researchers have used for genetics research.

The first "Biobank"

- Tissue samples (e.g., DNA) from 500,000 Britons
- Large-scale population study started in 2004, to investigate biological + genetic basis of low-frequency diseases
- Includes surveys for social outcomes, including education, occupation, etc.

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My access has not been processed yet, so I am merely describing the research design for using UK Biobank data.
$\rightarrow$ UK Biobank has a $N \approx 30,000$ subsample of people with genetic data from parents included; public data do not have this.

Data: ~~UK Biobank~~ Health & Retirement Study

A representative panel of retirement-age Americans 1990–2021.

$N \approx 500,000$ household heads with data on labour outcomes, age 50–65.[1]

[1] Limited to race = white, for whom Ed PGI is defined.

Data: Health & Retirement Study (HRS)

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household heads with data on labour outcomes, age 50–65.[2]

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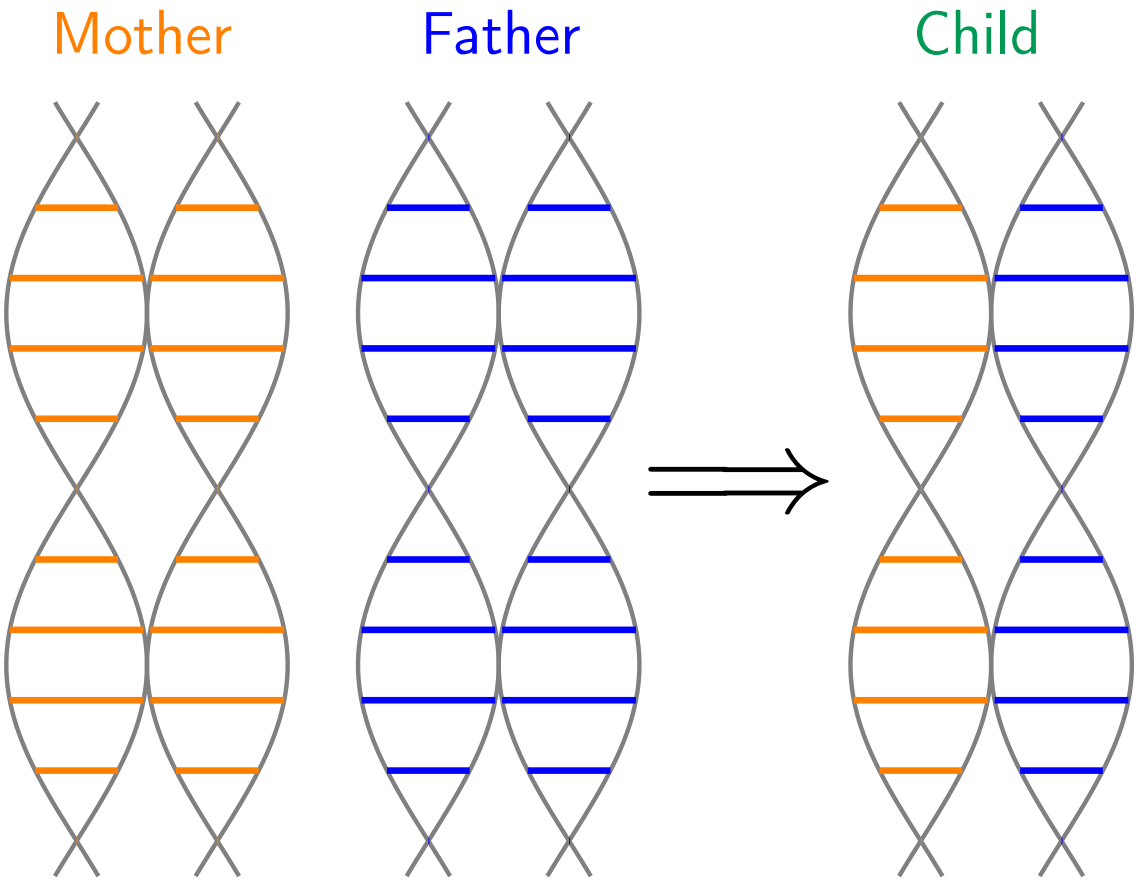
Genetic Associations: Independent Assortment

Question: Is Ed PGI correlated with demographics?

Genetic Associations: Independent Assortment

Question: Is Ed PGI randomly assigned?



Answer:
No, inherited from parents.

$$\mathbb{E}\left[Z_i^{\text{Child}} \,\middle|\, Z_i^{\text{Mother}}, Z_i^{\text{Father}}\right]$$

$$= \frac{1}{2}\left(Z_i^{\text{Mother}} + Z_i^{\text{Father}}\right)$$

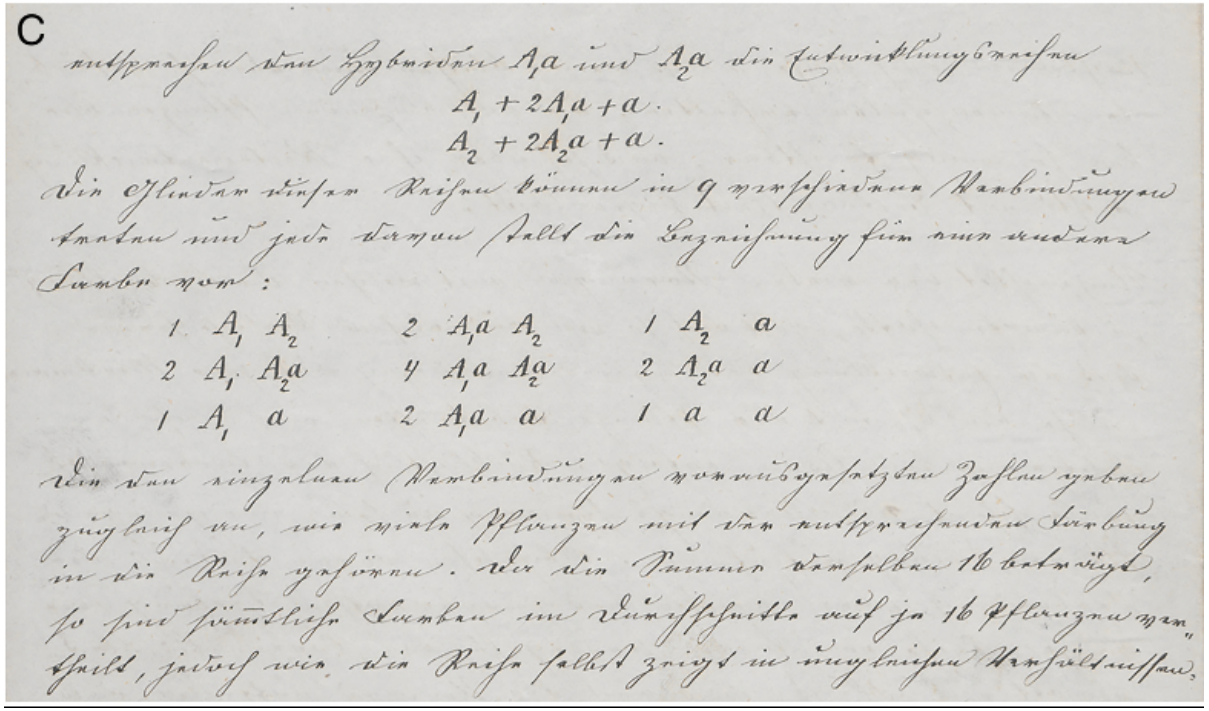
Genetic Associations: Independent Assortment

Difference in Child's Ed PGI, away from parents, arises from random inheritance + mixing, Mendelian Independent Assortment.

Gregor Mendel (1886) worked through principles of genetic inheritance (Stenseth Andersson Hoekstra, 2022 PNAS).

Genetic Associations: Independent Assortment

Difference in Ed PGI from parents' arises from random inheritance + mixing, Mendelian Independent Assortment.


C

entsprechen den Hybriden A_1a und A_2a die Entwicklungsreihen

$$A_1 + 2A_1a + a.$$
$$A_2 + 2A_2a + a.$$

Die Glieder dieser Reihen können in 9 verschiedenen Verbindungen treten und jede derselben stellt die Befruchtung für eine andere Samenpflanze vor:

1 A_1 A_2	2 A_1a A_2	1 A_2 a
2 A_1 A_2a	4 A_1a A_2a	2 A_2a a
1 A_1 a	2 A_1a a	1 a a

Die den einzelnen Verbindungen vorausgesetzten Zahlen geben zugleich an, wie viele Pflanzen mit den entsprechenden Färbung in die Reihe gehören. Wer die Summe derselben 16 beträgt, so sind künftighin Samen im Durchschnitte auf je 16 Pflanzen vertheilt, jedoch wie die Reihe selbst zeigt in ungleichen Verhältnissen.

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Difference in Ed PGI from parents' arises from random inheritance + mixing, Mendelian Independent Assortment.

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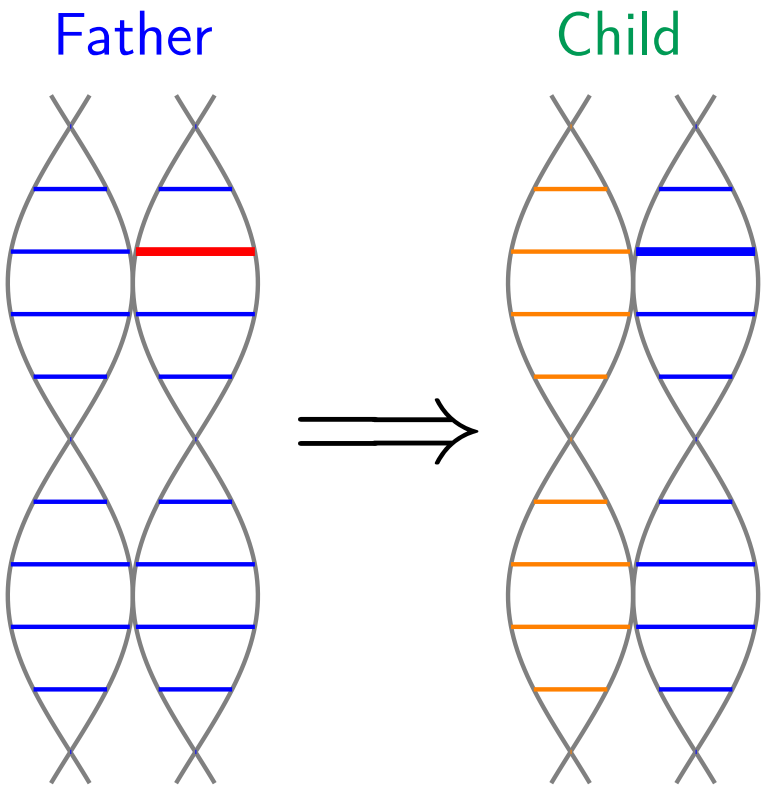

$$\Pr\left(\text{Child inherits} \,\middle|\, \text{Father one copy}\right) = \frac{1}{2}$$

I.e., inheritance prop. score depends only on parents' genetics.

Propagate this reasoning across all 1,271 single-letter changes in the Ed PGI, and both parents:

$$Z_i^{\text{Child}} - \mathbb{E}\left[Z_i^{\text{Child}} \,\middle|\, Z_i^{\text{Father}}, Z_i^{\text{Mother}}\right] \longrightarrow \text{Independently assigned.}$$

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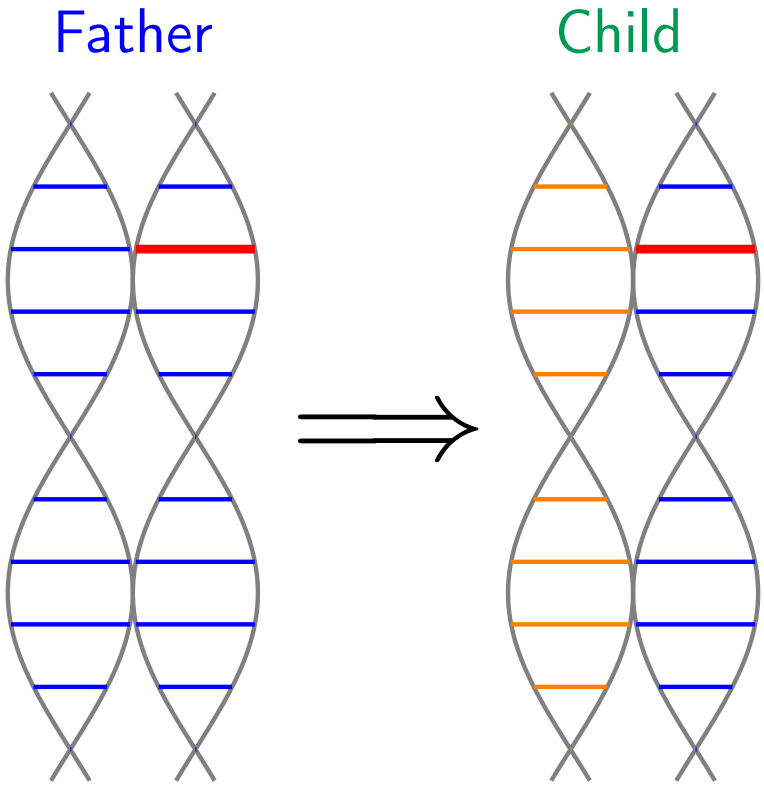

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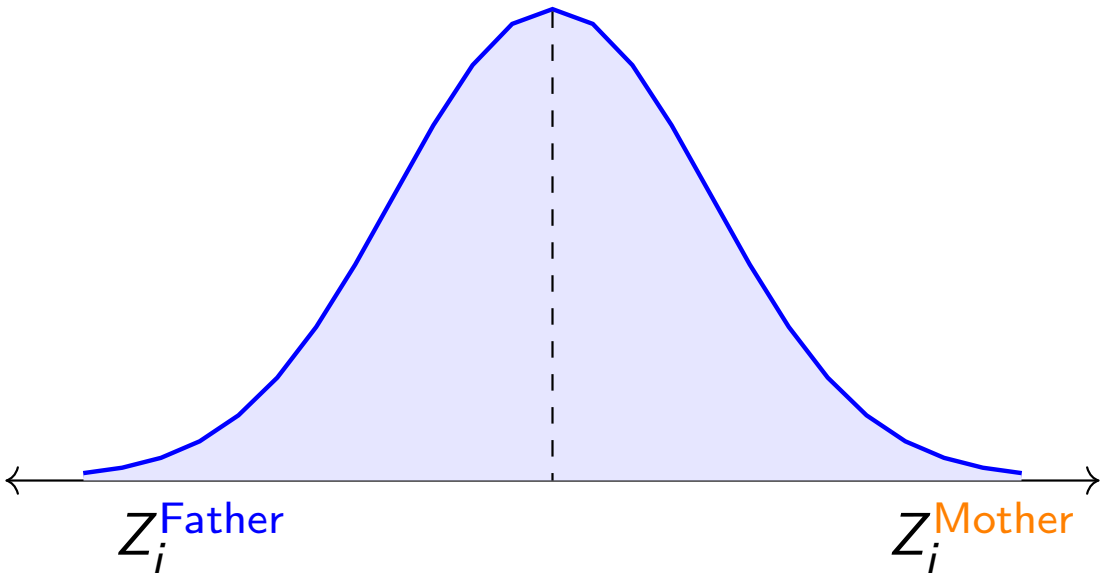

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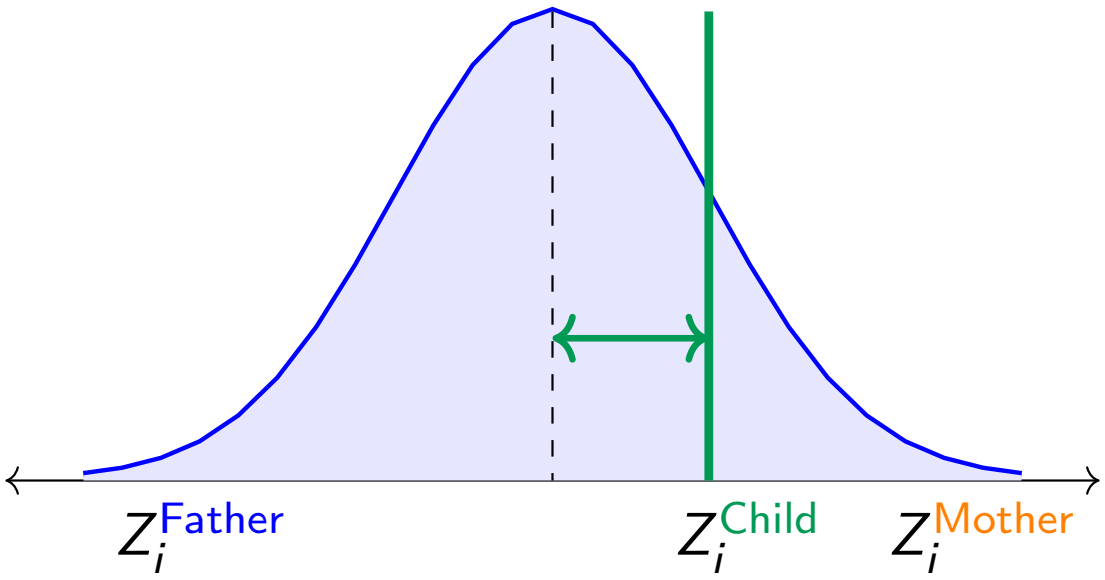

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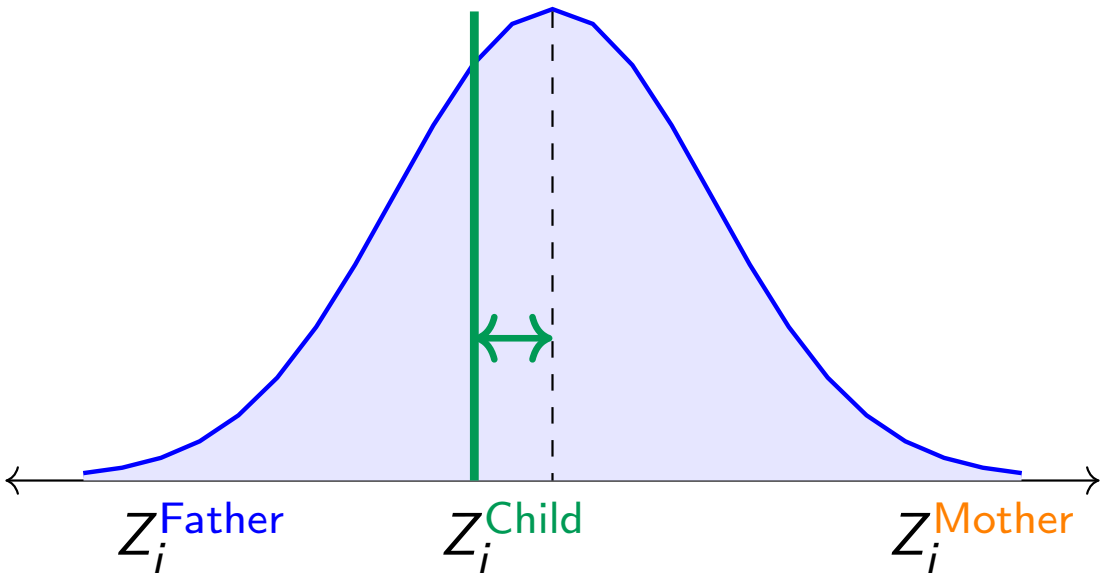

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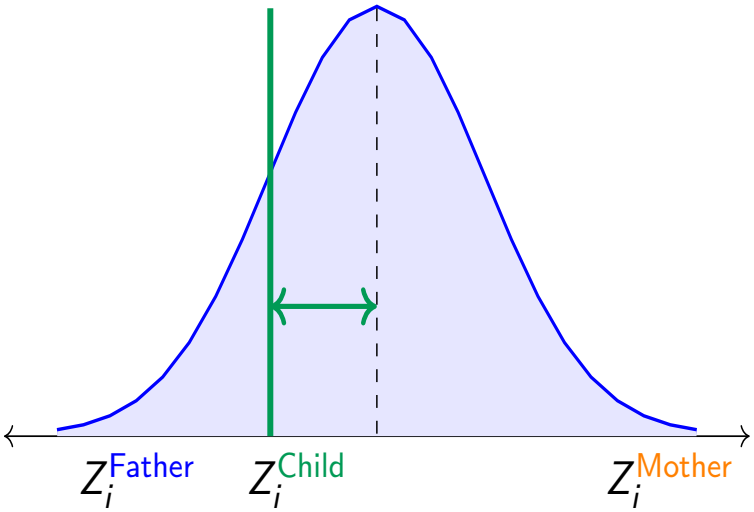

$\Longrightarrow$ Use the difference from parents as plausibly random genetic variation.

$$Z_i^{\text{Child}} - \mathbb{E}\left[Z_i^{\text{Child}} \,\middle|\, Z_i^{\text{Mother}}, Z_i^{\text{Father}}\right]$$

Remark on Identification:
Equivalent to instrumenting Child's Ed PGI using deviation from parents (e.g., formula instrument).

Genetic Associations: Outcomes

Question: Is Ed PGI related to labour market outcomes?

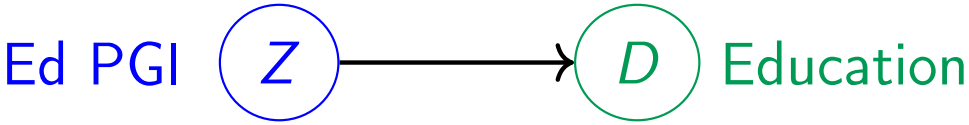

Figure: Bin Scatter, Ed PGI + Education Years.

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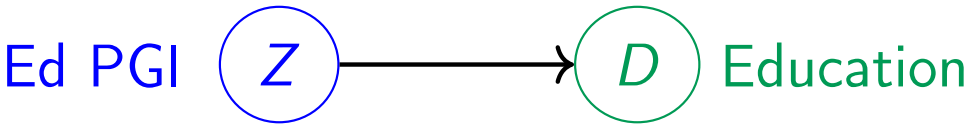

Figure: Bin Scatter Between Ed PGI + Education Years.

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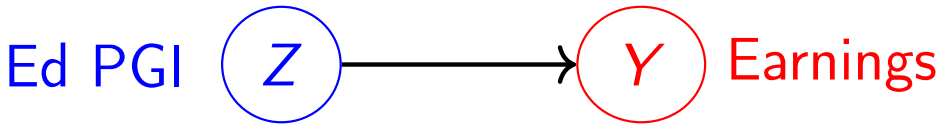

Figure: Bin Scatter Between Ed PGI + Annual Earnings.

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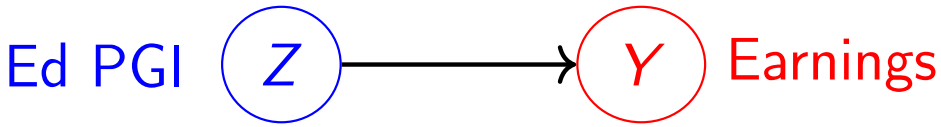
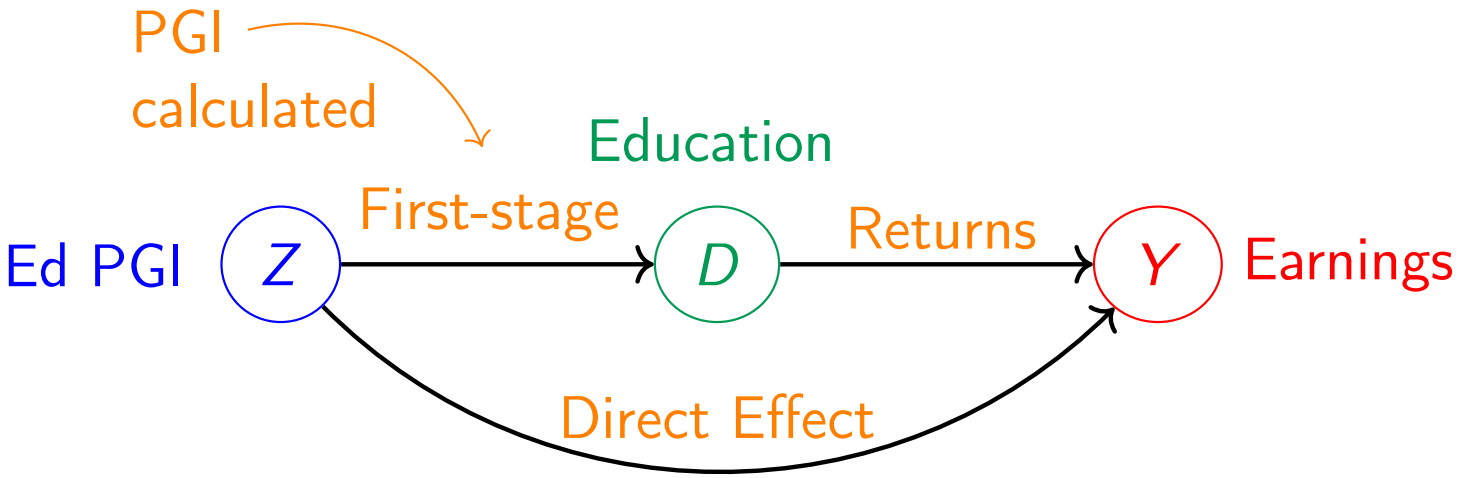


Figure: Bin Scatter Between Ed PGI + Annual Earnings.

Direct and Indirect Effects

Ed PGI is correlated with **education and income**, so what?
→ Ed PGI computed from training data on **only genes** + **education**.

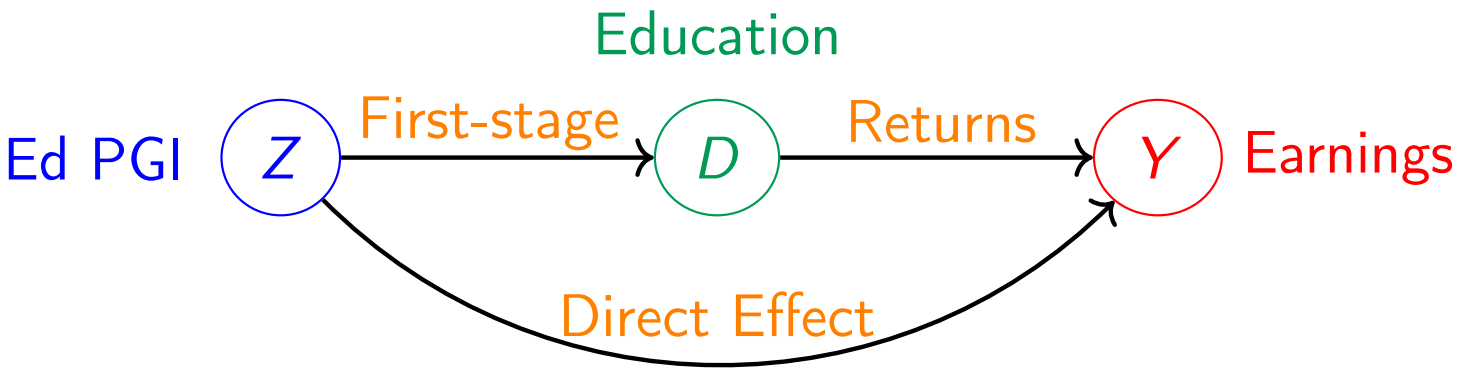


⟹ Association with earnings is non-trivial, can operate two channels:

Direct Genetic Channel	Returns to Education
▶ Creativity + Soft skills	▶ Classroom learning
▶ Correlation with first-stage	▶ Hard skills
▶ Signalling(?)	▶ Occupational qualifications

Direct and Indirect Effects

Mediation methods are a general-purpose tool to decompose total treatment effect: Direct Effect + Indirect effects.



$$\text{Direct Effect} = \underbrace{\mathbb{E}\left[Y_i \mid Z_i = 1, D_i\right] - \mathbb{E}\left[Y_i \mid Z_i = 0, D_i\right]}_{\text{Effect } Z \to Y\text{, holding } D \text{ constant}}$$

$$\text{Indirect Effect} = \text{First-stage} \times \underbrace{\left(\mathbb{E}\left[Y_i \mid Z_i, D_i = 1\right] - \mathbb{E}\left[Y_i \mid Z_i, D_i = 0\right]\right)}_{\text{Returns to Education, Estimate}}$$

Can be estimated with standard tools — e.g., simultaneous OLS, or anything (DML, etc.). ▸ Mediation, OLS

Direct and Indirect Effects: Selection Bias

Side-quest: Why should we not use mediation methods off-the-shelf?

- Mediation methods assume Education, D_i, is also randomly assigned.
- Major reason why these methods not used in labour/natural experiment research, despite popularity in other fields.

What are you getting, then? Bias Equations

$$\text{Estimand, Direct} = \text{Direct Effect} + \left(\text{Selection Bias} + \text{Complier Bias}\right)$$

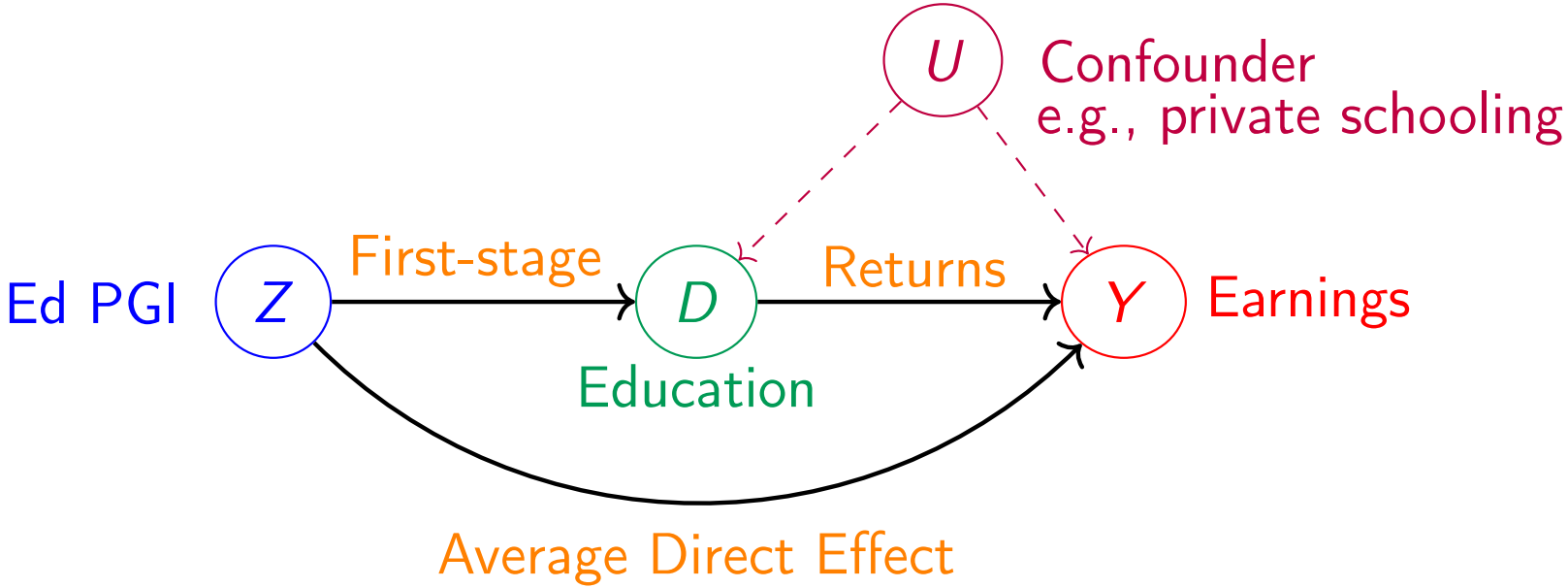
$$\text{Estimand, Indirect} = \text{Indirect Effect} + \left(\text{Selection Bias} + \text{Non-complier Bia}\right.$$

Bias terms persist, regardless of how random Ed PGI, Z_i is, with any selection-on-observables estimand (i.e., negative identification result).

Direct and Indirect Effects: Selection Bias

Side-quest: Why should we not use mediation methods off-the-shelf?

Figure: Selection Bias from Unobserved Confounder.



UKB has no info on private schooling...

- Selection bias from not controlling for private schooling
- Overestimate education returns → biased direct + indirect estimates.